



Certified Penetration Testing Professional

Wireshark Filters Cheat Sheet

Wireshark

Source: <https://www.wireshark.org>

Wireshark is a protocol analyzer tool that can capture and interactively browse the traffic running on a computer network. This tool uses Winpcap to capture packets on its supported networks. It captures live network traffic from Ethernet, IEEE 802.11, PPP/HDLC, ATM, Bluetooth, USB, Token Ring, Frame Relay, FDDI networks. The captured files can be programmatically edited via the command-line. A set of filters for customized data displays can be refined using a display filter.

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1. Frame Filters

Wireshark Filters	Description
<code>frame.cap_len</code>	Frame length stored into the capture file
<code>frame.coloring_rule.name</code>	Coloring Rule Name
<code>frame.coloring_rule.string</code>	Coloring Rule String
<code>frame.file_off</code>	File Offset
<code>frame.number</code>	Frame Number
<code>frame.p2p_dir</code>	Point-to-Point Direction
<code>frame.protocols</code>	Protocols in frame
<code>frame.ref_time</code>	Time Reference frame
<code>frame.time</code>	Arrival Time
<code>frame.ignored</code>	Frame is ignored
<code>frame.len</code>	Frame length on the wire
<code>frame.link_nr</code>	Link Number
<code>frame.marked</code>	Frame is marked
<code>frame.md5_hash</code>	Frame MD5 Hash
<code>frame.time_delta</code>	Time delta from previous captured frame
<code>frame.time_delta_displayed</code>	Time delta from previous displayed frame
<code>frame.time_epoch</code>	Epoch Time
<code>frame.time_invalid</code>	Arrival Time: Fractional second out of range (0-1000000000)
<code>frame.time_relative</code>	Time since reference or first frame

2. Ethernet Filters

Wireshark Filters	Description
<code>eth.addr</code>	Address of Ethernet or other MAC address
<code>eth.dst</code>	Destination of Ethernet or other MAC address
<code>eth.ig</code>	IG (Individual/Group) bit
<code>eth.len</code>	Length
<code>eth.lg</code>	LG (Local/Global) bit
<code>eth.src</code>	Source of Ethernet or other MAC address
<code>eth.trailer</code>	Trailer
<code>eth.type</code>	Type
<code>eth.src.oui_resolved</code>	Source OUI (Organizationally Unique Identifier) resolved
<code>eth.vlan.id</code>	VLAN
<code>eth.fcs_bad</code>	Bad checksum

3. DNS Filters

Wireshark Filters	Description
<code>dns.afsdb.hostname</code>	Display DNS AFSDb hostname
<code>dns.a</code>	Display IP address of DNS server
<code>dns.caa.tag_length</code>	Tag length
<code>dns.csync.flags</code>	Flags
<code>dns.count.prerequisites</code>	Prerequisites

4. ARP Filters

Wireshark Filters	Description
<code>arp.duplicate-address-detected</code>	Locate duplicate IP address traffic
<code>arp.proto.size</code>	Display protocol size
<code>arp.proto.type</code>	Display protocol type
<code>arp.src.hw_mac</code>	Display sender's MAC address
<code>arp.src.proto</code>	Display sender's protocol address
<code>arp.src.proto_ipv4</code>	Display sender's IP address
<code>arp.dst.hw_mac</code>	Target MAC address
<code>arp.dst.proto_ipv4</code>	Target IP address
<code>arp.hw.size</code>	Hardware size
<code>arp.hw.type</code>	Hardware type
<code>arp.opcode</code>	Opcode

5. PPP Filters

Wireshark Filters	Description
<code>ppp.address</code>	Address
<code>ppp.control</code>	Control
<code>ppp.direction</code>	Direction
<code>ppp.protocol</code>	Protocol

6. DTP Filters

Wireshark Filters	Description
<code>dtp.neighbor</code>	Neighbor
<code>dtp.tlv_type</code>	Type
<code>dtp.tlv_len</code>	Length
<code>dtp.version</code>	Version
<code>dtp.domain</code>	Domain
<code>dtp.senderid</code>	Sender ID
<code>dtp.tos</code>	Trunk Operating Status
<code>dtp.truncated</code>	DTP message is truncated prematurely
<code>dtp.data</code>	Data
<code>dtp.tas</code>	Trunk Administrative Status
<code>dtp.tat</code>	Trunk Administrative Type

7. IEEE 802.1Q VLAN Filters

Wireshark Filters	Description
<code>vlan.trailer</code>	Trailer
<code>vlan.priority</code>	Priority
<code>vlan.len</code>	Length
<code>vlan.cfi</code>	CFI
<code>vlan.etype</code>	Type
<code>vlan.id</code>	ID

<code>vlan.too_many_tags</code>	Too many nested VLAN tags
<code>vlan.len.past_end</code>	Length field value goes past the end of the payload

8. MPLS Filters

Wireshark Filters	Description
<code>mpls.bottom</code>	MPLS Bottom of Label Stack
<code>mpls.cw.control</code>	MPLS Control Channel
<code>mpls.cw.res</code>	MPLS Control Word Reserved
<code>mpls.exp</code>	MPLS Experimental Bits
<code>mpls.label</code>	MPLS Label
<code>mpls.oam.bip16</code>	BIP16
<code>mpls.oam.defect_location</code>	Defect Location (AS)
<code>mpls.oam.defect_type</code>	Defect Type
<code>mpls.oam.frequency</code>	Frequency
<code>mpls.oam.function_type</code>	Function Type
<code>mpls.oam.ttsi</code>	Trail Termination Source Identifier
<code>mpls.ttl</code>	MPLS TTL

9. VTP (VLAN Trunking Protocol) Filters

Wireshark Filters	Description
<code>vtp.code</code>	Code
<code>vtp.conf_rev_num</code>	Configuration Revision Number

vtp.followers	Followers
vtp.md	Management Domain
vtp.md5_digest	MD5 Digest
vtp.md_len	Management Domain Length
vtp.neighbor	Neighbor
vtp.seq_num	Sequence Number
vtp.start_value	Start Value
vtp.upd_id	Updater Identity
vtp.upd_ts	Update Timestamp
vtp.version	Version
vtp.vlan_info.802_10_index	802.10 Index
vtp.vlan_info.isl_vlan_id	ISL VLAN ID
vtp.vlan_info.len	VLAN Information Length
vtp.vlan_info.mtu_size	MTU Size
vtp.vlan_info.status.vlan_susp	VLAN suspended
vtp.vlan_info.tlv_len	Length
vtp.vlan_info.tlv_type	Type
vtp.vlan_info.vlan_name	VLAN Name
vtp.vlan_info.vlan_name_len	VLAN Name Length
vtp.vlan_info.vlan_type	VLAN Type

10. FR (Frame Relay) Filters

Wireshark Filters	Description
<code>fr.becn</code>	BECN
<code>fr.chdlctype</code>	Type
<code>fr.control</code>	Control Field
<code>fr.control.f</code>	Final
<code>fr.control.ftype</code>	Frame type
<code>fr.control.p</code>	Poll
<code>fr.control.s_ftype</code>	Supervisory frame type
<code>fr.control.u_modifier_cmd</code>	Command
<code>fr.control.u_modifier_resp</code>	Response
<code>fr.upper_dlci</code>	Upper Data Link Connection Identifier (DLCI)
<code>fr.third_dlci</code>	Third DLCI
<code>fr.snaptype</code>	Type
<code>fr.snap.pid</code>	Protocol ID
<code>fr.snap.oui</code>	Organization Code
<code>fr.second_dlci</code>	Second DLCI
<code>fr.lower_dlci</code>	Lower DLCI
<code>fr.dlcore_control</code>	DL-CORE Control

11. IPv4 Filters

Wireshark Filters	Description
<code>ip.addr</code>	Filter by Single IP Address displays all traffic from IP, be it source or destination
<code>ip.dst</code>	Filter by destination display traffic only from IP destination
<code>ip.addr== <IP address> && tcp.port== <port number></code>	Monitoring the Specific Port using the IP address and TCP port number.
<code>ip.addr == <IP address1> or ip.addr == <IP address2></code>	Filter by Multiple IP Addresses
<code>ip.checksum.status</code>	Display the header checksum status
<code>ip.src != xxx.xxx.xxx.xxx && ip.dst != xxx.xxx.xxx.xxx && sip</code>	Filter by a protocol (e.g. SIP) and filter out unwanted IPs
<code>ip.src</code>	Filter by Source IP display traffic only from IP source
<code>ip.checksum</code>	Header checksum
<code>ip.checksum_bad</code>	Bad
<code>ip.checksum_good</code>	Good
<code>ip.dsfield</code>	Differentiated Services Field
<code>ip.dsfield.ect</code>	ECN-Capable Transport (ECT)
<code>ip.dst</code>	Destination
<code>ip.dst_host</code>	Destination Host
<code>ip.flags</code>	Flags
<code>ip.flags.df</code>	Do not fragment
<code>ip.flags.mf</code>	More fragments
<code>ip.flags.rb</code>	Reserved bit
<code>ip.frag_offset</code>	Fragment offset

<code>ip.fragment.error</code>	Defragmentation error
<code>ip.fragment.multipletails</code>	Multiple tail fragments found
<code>ip.fragment.overlap</code>	Fragment overlap
<code>ip.fragment.overlap.conflict</code>	Conflicting data in fragment overlap
<code>ip.fragment.toolongfragment</code>	Fragment too long
<code>ip.fragments</code>	IPv4 Fragments
<code>ip.hdr_len</code>	Header Length
<code>ip.host</code>	Source or Destination Host
<code>ip.id</code>	Identification
<code>ip.len</code>	Total Length
<code>ip.proto</code>	Protocol
<code>ip.reassembled_in</code>	Reassembled IPv4 in frame
<code>ip.src</code>	Filter by Source IPv4
<code>ip.src_host</code>	Source Host
<code>ip.tos</code>	Type of Service
<code>ip.tos.cost</code>	Cost
<code>ip.tos.delay</code>	Delay
<code>ip.tos.precedence</code>	Precedence
<code>ip.tos.reliability</code>	Reliability
<code>ip.tos.throughput</code>	Throughput
<code>ip.ttl</code>	Time to live
<code>ip.version</code>	Version

12. IPv6 Filters

Wireshark Filters	Description
<code>ipv6.addr</code>	Source or Destination Address
<code>ipv6.class</code>	Traffic class
<code>ipv6.dst</code>	Destination
<code>ipv6.dst_host</code>	Destination Host
<code>ipv6.dst_opt</code>	Destination Option
<code>ipv6.flow</code>	Flow Label
<code>ipv6.fragment</code>	IPv6 Fragment
<code>ipv6.fragment.error</code>	Defragmentation error
<code>ipv6.hop_opt</code>	Hop-by-Hop Option
<code>ipv6.host</code>	Source or Destination Host
<code>ipv6.mipv6_home_address</code>	Home Address
<code>ipv6.mipv6_length</code>	Option Length
<code>ipv6.mipv6_type</code>	Option Type
<code>ipv6.nxt</code>	Next Header
<code>ipv6.fragment.more</code>	More Fragment
<code>ipv6.fragment.multipletails</code>	Multiple tail fragments found
<code>ipv6.fragment.offset</code>	Offset
<code>ipv6.fragment.overlap</code>	Fragment overlap
<code>ipv6.fragment.overlap.conflict</code>	Conflicting data in fragment overlap

<code>ipv6.fragment.toolongfragment</code>	Fragment too long
<code>ipv6.fragments</code>	IPv6 Fragments
<code>ipv6.fragment.id</code>	Identification
<code>ipv6.hlim</code>	Hop Limit
<code>ipv6.plen</code>	Payload Length
<code>ipv6.reassembled_in</code>	Reassembled IPv6 in frame
<code>ipv6.routing_hdr</code>	Routing Header, Type
<code>ipv6.routing_hdr.addr</code>	Address
<code>ipv6.routing_hdr.left</code>	Segments Left
<code>ipv6.routing_hdr.type</code>	Type
<code>ipv6.src_host</code>	Source Host
<code>ipv6.version</code>	Version

13. ICMP Filters

Wireshark Filters	Description
<code>icmp.checksum_bad</code>	Display bad checksum
<code>icmp.type==8 or icmp.type==0</code>	Detect an ICMP ping sweep attempt
<code>icmp.checksum</code>	Display checksum
<code>icmp.ident</code>	Display identifier (BE)
<code>icmp.seq</code>	Display sequence number (BE)
<code>icmp.mtu</code>	Display MTU of next hop
<code>icmp.redir_gw</code>	Display Gateway address

<code>icmp.checksum</code>	Checksum
<code>icmp.code</code>	Code
<code>icmp.type</code>	Type

14. ICMPv6 Filters

Wireshark Filters	Description
<code>icmpv6.all_comp</code>	All components
<code>icmpv6.checksum</code>	Checksum
<code>icmpv6.checksum_bad</code>	Bad checksum
<code>icmpv6.code</code>	Code
<code>icmpv6.comp</code>	Component
<code>icmpv6.haad.ha_addrs</code>	Home agent addresses
<code>icmpv6.identifier</code>	Identifier
<code>icmpv6.option</code>	ICMPv6 Option
<code>icmpv6.option.cga</code>	CGA
<code>icmpv6.option.length</code>	Length
<code>icmpv6.option.name_type</code>	Name type
<code>icmpv6.option.name_type.fqdn</code>	FQDN
<code>icmpv6.option.name_x501</code>	DER Encoded X.501 Name
<code>icmpv6.option.rsa.key_hash</code>	Key hash
<code>icmpv6.option.type</code>	Type
<code>icmpv6.ra.cur_hop_limit</code>	Cur hop limit

<code>icmpv6.ra.reachable_time</code>	Reachable time
<code>icmpv6.ra.retrans_timer</code>	Retrans timer
<code>icmpv6.ra.router_lifetime</code>	Router lifetime
<code>icmpv6.recursive_dns_serv</code>	Recursive DNS Servers
<code>icmpv6.type</code>	Type

15. TCP Filters

Wireshark Filters	Description
<code>tcp contains traffic</code>	Displays all TCP packets that contains the word 'traffic'
<code>tcp.port==23</code>	Monitoring the Specific Port using tcp port number.
<code>tcp.flags.reset==1</code>	Displays all TCP resets
<code>tcp.analysis.retransmission</code>	Displays all retransmissions in the trace
<code>tcp.port == 4000</code>	Sets a filter for any TCP packet with 4000 as a source or destination port
<code>tcp.port eq 25 or icmp</code>	Displays only SMTP (port 25) and ICMP traffic
<code>tcp.dstport==7</code>	Detect TCP ping sweep attempt
<code>tcp.flags==0X029</code>	View the packets with FIN, PSH, and URG TCP flags set
<code>tcp.ack</code>	Acknowledgment number
<code>tcp.checksum</code>	Checksum
<code>tcp.checksum_bad</code>	Bad Checksum

<code>tcp.checksum_good</code>	Good Checksum
<code>tcp.options.qs</code>	TCP QS Option
<code>tcp.options.sack</code>	TCP SACK Option
<code>tcp.options.sack_le</code>	TCP SACK Left Edge
<code>tcp.options.sack_perm</code>	TCP SACK Permitted Option
<code>tcp.continuation_to</code>	This is a continuation to the PDU in frame
<code>tcp.options.sack_re</code>	TCP SACK Right Edge
<code>tcp.options.time_stamp</code>	TCP Time Stamp Option
<code>tcp.flags</code>	Flags
<code>tcp.dstport</code>	Destination Port
<code>tcp.options.wscale</code>	TCP Window Scale Option
<code>tcp.flags.ack</code>	Acknowledgment
<code>tcp.options.wscale_val</code>	TCP Windows Scale Option Value
<code>tcp.flags.cwr</code>	Congestion Window Reduced (CWR)
<code>tcp.pdu.last_frame</code>	Last frame of this PDU
<code>tcp.flags.ecn</code>	ECN-Echo
<code>tcp.pdu.size</code>	PDU Size
<code>tcp.flags.fin</code>	Fin
<code>tcp.pdu.time</code>	Time until the last segment of this PDU
<code>tcp.flags.push</code>	Push
<code>tcp.port</code>	Source or Destination Port

<code>tcp.flags.reset</code>	Reset
<code>tcp.flags.syn</code>	TCP; Filter SYN flag
<code>tcp.flags.urg</code>	Urgent
<code>tcp.hdr_len</code>	Header Length
<code>tcp.reassembled_in</code>	Reassembled PDU in frame
<code>tcp.segment</code>	TCP Segment
<code>tcp.segment.error</code>	Reassembling error
<code>tcp.segment.multipletails</code>	Multiple tail segments found
<code>tcp.len</code>	TCP Segment Len
<code>tcp.nxtseq</code>	Next sequence number
<code>tcp.options</code>	TCP Options
<code>tcp.options.cc</code>	TCP CC Option
<code>tcp.options.ccecho</code>	TCP CC Echo Option
<code>tcp.options.ccnew</code>	TCP CC New Option
<code>tcp.options.echo</code>	TCP Echo Option
<code>tcp.options.echo_reply</code>	TCP Echo Reply Option
<code>tcp.options.md5</code>	TCP MD5 Option
<code>tcp.options.mss</code>	TCP MSS Option
<code>tcp.options.mss_val</code>	MSS Value
<code>tcp.segment.overlap</code>	Segment overlap
<code>tcp.segment.overlap.conflict</code>	Conflicting data in segment overlap

<code>tcp.segment.toolongfragment</code>	Segment too long
<code>tcp.segments</code>	Reassembled TCP Segments
<code>tcp.seq</code>	Sequence number
<code>tcp.srcport</code>	Source Port
<code>tcp.time_delta</code>	Time since previous frame in this TCP stream
<code>tcp.time_relative</code>	Time since first frame in this TCP stream
<code>tcp.urgent_pointer</code>	Urgent pointer
<code>tcp.window_size</code>	Calculated window size

16. UDP Filters

Wireshark Filters	Description
<code>udp contains 33:27:58</code>	Sets a filter for the HEX values of 0x33 0x27 0x58 at any offset
<code>udp.dstport==7</code>	Detect UDP ping sweep attempt
<code>udp.proc.dstuid</code>	Displays the destination process User ID
<code>udp.proc.srcuid</code>	Displays the source process User ID
<code>udp.checksum</code>	Checksum
<code>udp.dstport</code>	Destination Port
<code>udp.srcport</code>	Source Port
<code>udp.checksum_bad</code>	Bad checksum
<code>udp.length</code>	Length
<code>udp.checksum.zero</code>	Illegal Checksum value (0)
<code>udp.pdu.size</code>	PDU Size

<code>udp.possible_traceroute</code>	Possible traceroute
<code>udp.proc.dstpid</code>	Destination process ID
<code>udp.stream</code>	Stream index
<code>udp.time_delta</code>	Time since previous frame
<code>udp.time_relative</code>	Time since first frame
<code>udplite.checksum_coverage.bad</code>	Bad checksum coverage length value

17. HTTP Filters

Wireshark Filters	Description
<code>http.request</code>	Displays and filter all HTTP GET requests
<code>http.accept</code>	Accept
<code>http.accept_encoding</code>	Accept Encoding
<code>http.accept_language</code>	Accept-Language
<code>http.authbasic</code>	Credentials
<code>http.authcitrix</code>	Citrix AG Auth
<code>http.authcitrix.domain</code>	Citrix AG Domain
<code>http.authcitrix.password</code>	Citrix AG Password
<code>http.authcitrix.session</code>	Citrix AG Session ID
<code>http.authcitrix.user</code>	Citrix AG Username
<code>http.authorization</code>	Authorization
<code>http.referer</code>	Referer
<code>http.cache_control</code>	Cache-Control

<code>http.connection</code>	Connection
<code>http.content_encoding</code>	Content-Encoding
<code>http.content_length</code>	Content length
<code>http.content_type</code>	Content-Type
<code>http.cookie</code>	Cookie
<code>http.date</code>	Date
<code>http.file_data</code>	File Data
<code>http.host</code>	Host
<code>http.last_modified</code>	Last-Modified
<code>http.location</code>	Location
<code>http.notification</code>	Notification
<code>http.proxy_authenticate</code>	Proxy-Authenticate
<code>http.request.method</code>	Request Method
<code>http.request.uri</code>	Request URI
<code>http.request.version</code>	Request Version
<code>http.response</code>	Filter all HTTP get responses
<code>http.response.code</code>	HTTP
<code>http.request or http.response</code>	Filter all HTTP get requests and responses
<code>http.server</code>	Server
<code>http.set_cookie</code>	Set-Cookie
<code>http.transfer_encoding</code>	Transfer-Encoding
<code>http.user_agent</code>	User-Agent

<code>http.www_authenticate</code>	WWW-Authenticate
<code>http.x_forwarded_for</code>	X-Forwarded-For
<code>!http.user_agent contains "Firefox" && !http.user_agent contains "Chrome"</code>	Find user agents
<code>frame contains "GET" && !tcp.port==80</code>	HTTP Get not on port 80
<code>http.response.code > 299 && http.response.code < 400</code>	HTTP Redirections
<code>http.request.uri matches "\.(exe zip jar)\$"</code>	HTTP .exe,.zip,.jar objects
<code>http.request.method in {PUT POST}</code>	HTTP PUT and POST messages

18. BGP Filters

Wireshark Filters	Description
<code>bgp.aggregator_as</code>	Aggregator AS
<code>bgp.aggregator_origin</code>	Aggregator origin
<code>bgp.as_path</code>	AS Path
<code>bgp.cluster_identifier</code>	Cluster identifier
<code>bgp.cluster_list</code>	Cluster List
<code>bgp.community_as</code>	Community AS
<code>bgp.community_value</code>	Community value
<code>bgp.local_pref</code>	Local preference
<code>bgp.mp_nlri_tnl_id</code>	MP Reach NLRI Tunnel Identifier
<code>bgp.mp_reach_nlri_ipv4_prefix</code>	MP Reach NLRI IPv4 prefix
<code>bgp.mp_unreach_nlri_ipv4_prefix</code>	MP Unreach NLRI IPv4 prefix

<code>bgp.multi_exit_disc</code>	Multiple exit discriminator
<code>bgp.next_hop</code>	Next hop
<code>bgp.nlri_prefix</code>	NLRI prefix
<code>bgp.origin</code>	Origin
<code>bgp.originator_id</code>	Originator identifier
<code>bgp.type</code>	Type
<code>bgp.withdrawn_prefix</code>	Withdrawn prefix
<code>bgp.wildcard_route_target</code>	Wildcard route target
<code>bgp.vplsbgp.labelblock.size</code>	Label Block Size
<code>bgp.vplsbgp.labelblock.offset</code>	Label Block Offset
<code>bgp.vplsbgp.labelblock.base</code>	Label Block Base
<code>bgp.vplsad.length</code>	Length
<code>bgp.update.withdrawn_routes.length</code>	Withdrawn Routes Length
<code>bgp.update.withdrawn_routes</code>	Withdrawn Routes
<code>bgp.update.path_attribute.multi_exit_disc</code>	Multiple exit discriminator
<code>bgp.update.path_attribute.next_hop</code>	Next hop
<code>bgp.unknown_safi</code>	Unknown SAFI
<code>bgp.unknown_label</code>	Unknown Label VPN
<code>bgp.unknown_afi</code>	Unknown Address Family
<code>bgp.ssa_value</code>	Value
<code>bgp.ssa_type</code>	SSA Type
<code>bgp.ssa_t</code>	Transitive bit

<code>bgp.ssa_12tpv3_Unused</code>	Unused
<code>bgp.ssa_12tpv3_session_id</code>	Session ID
<code>bgp.ssa_12tpv3_s</code>	Sequencing bit
<code>bgp.ssa_12tpv3_pref</code>	Preference
<code>bgp.ssa_12tpv3_cookie_len</code>	Cookie Length
<code>bgp.route_refresh.subtype</code>	Subtype
<code>bgp.route_refresh.orf.entry.ip</code>	ORFEntry IP address
<code>bgp.rd</code>	Route Distinguisher
<code>bgp.prefix_sid.type_err</code>	BGP Prefix-SID unknown TLV type
<code>bgp.prefix_length</code>	Prefix Length
<code>bgp.path_attribute.cluster_id</code>	Cluster ID
<code>bgp.path_attribute.cluster_list</code>	Cluster List
<code>bgp.open.version</code>	Version
<code>bgp.open.opt.param.type</code>	Parameter Type
<code>bgp.open.holdtime</code>	Hold Time
<code>bgp.notify.minor_error_update</code>	Minor error Code (Update Message)
<code>bgp.notify.minor_error_unknown</code>	Minor error Code (Unknown)
<code>bgp.notify.minor_error_state</code>	Minor error Code (State Machine)
<code>bgp.notify.minor_error_open</code>	Minor error Code (Open Message)
<code>bgp.nlri_path_id</code>	NLRI path id
<code>bgp.mp_unreach_nlri_ipv6_prefix</code>	MP Unreach NLRI IPv6 prefix

<code>bgp.mdt_safi_rd</code>	Route Distinguisher
<code>bgp.mdt_safi_ipv4_addr</code>	IPv4 Address
<code>bgp.mdt_safi_group_addr</code>	Group Address
<code>bgp.mcast_vpn_nlri_source_length</code>	Multicast Source Length
<code>bgp.marker_invalid</code>	Marker is not all ones
<code>bgp.marker</code>	Marker
<code>bgp.ls.warn</code>	Link State warning
<code>bgp.ls.tlv.unreserved_bandwidth</code>	Unreserved bandwidth TLV
<code>bgp.ls.node_flag_bits.abr</code>	ABR Bit
<code>bgp.ls.node_flag_bits.external</code>	External Bit
<code>bgp.ls.bandwidth_value</code>	Bandwidth
<code>bgp.ls.error</code>	Link State error
<code>bgp.large_communities</code>	Large Communities
<code>bgp.label_stack</code>	Label Stack
<code>bgp.flowspec_route_distinguisher</code>	Route Distinguisher
<code>bgp.flags.transitive</code>	Transitive
<code>bgp.ext_communities</code>	Carried extended communities
<code>bgp.evpn.nlri.rt</code>	Route Type
<code>bgp.cap.orf.sendreceive</code>	Send Receive
<code>bgp.attr.pmsi.tunnel_type</code>	Unknown Tunnel type
<code>bgp.afi_type_not_supported</code>	AFI Type not supported
<code>bgp.ad.pe_addr</code>	PE Addr

19. TLS Filters

Wireshark Filters	Description
<code>tls</code>	All TLS Packets
<code>tls.record.content_type == 22</code>	TLS Handshake Packets
<code>tls.handshake</code>	Handshake Protocol
<code>tls.handshake.cert_type</code>	Certificate Type
<code>tls.handshake.cert_type.type</code>	Certificate Type
<code>tls.handshake.cert_type.types</code>	Certificate Type List
<code>tls.handshake.cert_type.types_len</code>	Certificate Type List Length
<code>tls.handshake.type == 1</code>	TLS Client Hello Packets
<code>tls.handshake.type == 2</code>	TLS Server Hello Packets
<code>tls.record.content_type == 21</code>	TLS Encrypted Alert
<code>tls.handshake.cert_types</code>	Certificate types
<code>tls.handshake.cert_types_count</code>	Certificate types count
<code>tls.handshake.cert_url.padding</code>	Padding
<code>tls.handshake.cert_url.sha1</code>	SHA1 Hash
<code>tls.handshake.cert_url.url</code>	URL
<code>tls.handshake.cert_url.url_hash</code>	URL and Hash
<code>tls.handshake.cert_url.url_len</code>	URL Length
<code>tls.handshake.cert_url.url_hash_len</code>	URL and Hash list Length
<code>tls.handshake.cert_url_type</code>	Certificate Chain Type
<code>tls.handshake.certificate</code>	Certificate

<code>tls.handshake.certificate_length</code>	Certificate Length
<code>tls.handshake.certificate_request_context</code>	Certificate Request Context
<code>tls.handshake.certificate_request_context_length</code>	Certificate Request Context Length
<code>tls.handshake.certificates</code>	Certificates
<code>tls.handshake.certificates_length</code>	Certificates Length
<code>tls.handshake.challenge</code>	Challenge
<code>tls.handshake.challenge_length</code>	Challenge Length
<code>tls.handshake.cipher_spec_len</code>	Cipher Spec Length
<code>tls.handshake.cipher_suites_length</code>	Cipher Suites Length
<code>tls.handshake.cipherspec</code>	Cipher Spec
<code>tls.handshake.ciphersuite</code>	Cipher Suite
<code>tls.handshake.ciphersuites</code>	Cipher Suites
<code>tls.handshake.clear_key_data</code>	Clear Key Data
<code>tls.handshake.clear_key_length</code>	Clear Key Data Length
<code>tls.handshake.client_cert_vrfy.sig</code>	Signature
<code>tls.handshake.client_cert_vrfy.sig_len</code>	Signature length
<code>tls.handshake.client_point</code>	Pubkey
<code>tls.handshake.client_point_len</code>	Pubkey Length
<code>tls.handshake.comp_method</code>	Compression Method
<code>tls.handshake.extensions_server_name contains "hack"</code>	TLS contains "hack" in server name

<code>tls.alert_message</code>	Alert Message
<code>tls.alert_message.desc</code>	Description
<code>tls.alert_message.level</code>	Level
<code>tls.app_data</code>	Encrypted Application Data
<code>tls.change_cipher_spec</code>	Change Cipher Spec Message
<code>tls.compress_certificate.algorithm</code>	Algorithm
<code>tls.compress_certificate.algorithms_length</code>	Algorithms Length
<code>tls.compress_certificate.compressed_certificate_message</code>	Compressed Certificate Message
<code>tls.compress_certificate.compressed_certificate_message.length</code>	Length
<code>tls.compress_certificate.uncompressed_length</code>	Uncompressed Length
<code>tls.early_data.max_early_data_size</code>	Maximum Early Data Size
<code>tls.esni.encrypted_sni</code>	Encrypted SNI
<code>tls.esni.encrypted_sni_length</code>	Encrypted SNI Length
<code>tls.esni.nonce</code>	Nonce
<code>tls.esni.record_digest</code>	Record Digest
<code>tls.esni.record_digest_length</code>	Record Digest Length
<code>tls.esni.suite</code>	Cipher Suite
<code>tls.extension.oid_filters.oid</code>	Certificate Extension OID
<code>tls.extension.oid_filters.oid_length</code>	Certificate Extension OID Length
<code>tls.extension.oid_filters.values_length</code>	Certificate Extension Values Length

<code>tls.extension.oid_filters_length</code>	OID Filters Length
<code>tls.extension.psk_ke_mode</code>	PSK Key Exchange Mode
<code>tls.extension.psk_ke_modes_length</code>	PSK Key Exchange Modes Length
<code>tls.handshake.comp_methods</code>	Compression Methods
<code>tls.handshake.comp_methods_length</code>	Compression Methods Length
<code>tls.handshake.connection_id</code>	Connection ID
<code>tls.handshake.connection_id_length</code>	Connection ID Length
<code>tls.handshake.dname</code>	Distinguished Name
<code>tls.handshake.dname_len</code>	Distinguished Name Length
<code>tls.handshake.dnames</code>	Distinguished Names
<code>tls.handshake.dnames_len</code>	Distinguished Names Length
<code>tls.handshake.encrypted_key</code>	Encrypted Key
<code>tls.handshake.encrypted_key_length</code>	Encrypted Key Data Length
<code>tls.handshake.epms</code>	Encrypted PreMaster
<code>tls.handshake.epms_len</code>	Encrypted PreMaster length
<code>tls.handshake.exponent</code>	Exponent
<code>tls.handshake.exponent_len</code>	Exponent Length
<code>tls.handshake.extension.data</code>	Data
<code>tls.handshake.extension.heartbeat.mode</code>	Mode
<code>tls.handshake.extension.len</code>	Length
<code>tls.handshake.extension.type</code>	Type

<code>tls.handshake.extensions.cookie</code>	Cookie
<code>tls.handshake.extensions.cookie_len</code>	Cookie length
<code>tls.handshake.extensions.psk.binders</code>	PSK Binders
<code>tls.handshake.extensions.psk.binders_len</code>	PSK Binders length
<code>tls.handshake.extensions.psk.identities.length</code>	Identities Length
<code>tls.handshake.extensions.psk.identity.identity</code>	Identity
<code>tls.handshake.extensions.psk.identity.identity_length</code>	Identity Length
<code>tls.handshake.extensions.psk.identity.obfuscated_ticket_age</code>	Obfuscated Ticket Age
<code>tls.handshake.extensions.psk.identity.selected</code>	Selected Identity
<code>tls.handshake.extensions.supported_version</code>	Supported Version
<code>tls.handshake.extensions.supported_versions_len</code>	Supported Versions length
<code>tls.handshake.extensions_alpn_len</code>	ALPN Extension Length
<code>tls.handshake.extensions_alpn_list</code>	ALPN Protocol
<code>tls.handshake.extensions_alpn_str</code>	ALPN Next Protocol
<code>tls.handshake.extensions_alpn_str_len</code>	ALPN string length
<code>tls.handshake.extensions_ec_point_format</code>	EC point format
<code>tls.handshake.extensions_ec_point_formats</code>	EC point formats
<code>tls.handshake.extensions_ec_point_formats_length</code>	EC point formats Length

<code>tls.handshake.extensions_key_share_client_length</code>	Client Key Share Length
<code>tls.handshake.extensions_key_share_group</code>	Group
<code>tls.handshake.extensions_key_share_key_exchange</code>	Key Exchange
<code>tls.handshake.extensions_key_share_key_exchange_length</code>	Key Exchange Length
<code>tls.handshake.extensions_key_share_selected_group</code>	Selected Group
<code>tls.handshake.extensions_length</code>	Extensions Length
<code>tls.handshake.extensions_npn</code>	Next Protocol
<code>tls.handshake.extensions_npn_str_len</code>	Protocol string length
<code>tls.handshake.extensions_padding_data</code>	Padding Data
<code>tls.handshake.extensions_reneg_info</code>	Renegotiation info
<code>tls.handshake.extensions_reneg_info_len</code>	Renegotiation info extension length
<code>tls.handshake.extensions_server_name</code>	Server Name
<code>tls.handshake.extensions_server_name_len</code>	Server Name length
<code>tls.handshake.extensions_server_name_list_len</code>	Server Name list length
<code>tls.handshake.extensions_server_name_type</code>	Server Name Type
<code>tls.handshake.extensions_status_request_exts_len</code>	Request Extensions Length
<code>tls.handshake.extensions_status_request_len</code>	Certificate Status Length
<code>tls.handshake.extensions_status_request_list_len</code>	Certificate Status List Length

<code>tls.handshake.extensions_status_request_responder_ids_len</code>	Responder ID list Length
<code>tls.handshake.extensions_status_request_type</code>	Certificate Status Type
<code>tls.handshake.extensions_supported_group</code>	Supported Group
<code>tls.handshake.extensions_supported_groups</code>	Supported Groups List
<code>tls.handshake.extensions_supported_groups_length</code>	Supported Groups List Length
<code>tls.handshake.fragment</code>	Handshake Fragment
<code>tls.handshake.fragment.count</code>	Handshake Fragment count
<code>tls.handshake.fragments</code>	Reassembled Handshake Fragments
<code>tls.handshake.hint</code>	Hint
<code>tls.handshake.hint_len</code>	Hint Length
<code>tls.handshake.identity</code>	Identity
<code>tls.handshake.identity_len</code>	Identity Length
<code>tls.handshake.key_arg</code>	Key Argument
<code>tls.handshake.key_arg_length</code>	Key Argument Length
<code>tls.handshake.key_update.request_update</code>	Key Update Request
<code>tls.handshake.length</code>	Length
<code>tls.handshake.max_fragment_length</code>	Maximum Fragment Length
<code>tls.handshake.md5_hash</code>	MD5 Hash
<code>tls.handshake.modulus</code>	Modulus
<code>tls.handshake.modulus_len</code>	Modulus Length

<code>tls.handshake.npn_padding</code>	Padding
<code>tls.handshake.npn_padding_len</code>	Padding Length
<code>tls.handshake.npn_selected_protocol</code>	Selected Protocol
<code>tls.handshake.npn_selected_protocol_len</code>	Selected Protocol Length
<code>tls.handshake.ocsp_response_len</code>	OCSP Response Length
<code>tls.handshake.ocsp_response_list_len</code>	OCSP Response List Length
<code>tls.handshake.random</code>	Random
<code>tls.handshake.random_bytes</code>	Random Bytes
<code>tls.handshake.random_time</code>	GMT Unix Time
<code>tls.handshake.reassembled_in</code>	Reassembled Handshake Message in frame
<code>tls.handshake.server_curve_type</code>	Curve Type
<code>tls.handshake.server_named_curve</code>	Named Curve
<code>tls.handshake.server_point</code>	Pubkey
<code>tls.handshake.server_point_len</code>	Pubkey Length
<code>tls.handshake.session_id</code>	Session ID
<code>tls.handshake.session_id_hit</code>	Session ID Hit
<code>tls.handshake.session_id_length</code>	Session ID Length
<code>tls.handshake.session_id_length.error</code>	Session ID length error
<code>tls.handshake.session_ticket</code>	Session Ticket
<code>tls.handshake.session_ticket_age_add</code>	Session Ticket Age Add
<code>tls.handshake.session_ticket_length</code>	Session Ticket Length

<code>tls.handshake.session_ticket_lifetime_hint</code>	Session Ticket Lifetime Hint
<code>tls.handshake.session_ticket_nonce</code>	Session Ticket Nonce
<code>tls.handshake.session_ticket_nonce_length</code>	Session Ticket Nonce Length
<code>tls.handshake.status_request.undecoded</code>	Responder ID list or Request Extensions are not implemented, contact Wireshark developers if you want this to be supported
<code>tls.handshake.verify_data</code>	Verify Data
<code>tls.heartbeat_message</code>	Heartbeat Message
<code>tls.heartbeat_message.padding</code>	Payload Length
<code>tls.heartbeat_message.payload</code>	Payload Length
<code>tls.ignored_unknown_record</code>	Ignored Unknown Record
<code>tls.malformed.buffer_too_small</code>	Malformed message, not enough data is available
<code>tls.malformed.trailing_data</code>	Undecoded trailing data is present
<code>tls.malformed.vector_length</code>	The variable vector length is outside the permitted range
<code>tls.quic.initial_version</code>	Initial Version
<code>tls.quic.len</code>	Parameters Length
<code>tls.quic.negotiated_version</code>	Negotiated Version
<code>tls.quic.parameter</code>	Parameter
<code>tls.quic.parameter.ack_delay_exponent</code>	ack_delay_exponent
<code>tls.quic.parameter.active_connection_id_limit</code>	Active Connection ID Limit

<code>tls.quic.parameter.idle_timeout</code>	idle_timeout
<code>tls.quic.parameter.initial_max_data</code>	initial_max_data
<code>tls.reassembled.data</code>	Reassembled PDU data
<code>tls.reassembled.length</code>	Reassembled PDU length
<code>tls.reassembled_in</code>	Reassembled PDU in frame
<code>tls.record</code>	Record Layer
<code>tls.record.content_type</code>	Content Type
<code>tls.record.is_escape</code>	Is Escape
<code>tls.record.length</code>	Length
<code>tls.record.length.invalid</code>	Record fragment length is too small or too large
<code>tls.record.opaque_type</code>	Opaque Type
<code>tls.record.padding_length</code>	Padding Length
<code>tls.record.version</code>	Version
<code>tls.record_size_limit</code>	Record Size Limit
<code>tls.resumed</code>	This session reuses previously negotiated keys (Session resumption)
<code>tls.sct.sct_extensions</code>	Extensions
<code>tls.sct.sct_extensions_length</code>	Extensions length
<code>tls.sct.sct_length</code>	Serialized SCT Length
<code>tls.sct.sct_logid</code>	Log ID
<code>tls.sct.sct_signature</code>	Signature

<code>tls.sct.sct_signature_length</code>	Signature Length
<code>tls.sct.sct_timestamp</code>	Timestamp
<code>tls.sct.sct_version</code>	SCT Version
<code>tls.sct.scts_length</code>	Serialized SCT List Length
<code>tls.segment</code>	TLS segment
<code>tls.segment.count</code>	Segment count
<code>tls.segment.data</code>	TLS segment data
<code>tls.segment.error</code>	Reassembling error
<code>tls.segment.multipletails</code>	Multiple tail segments found
<code>tls.segment.overlap</code>	Segment overlap
<code>tls.segment.overlap.conflict</code>	Conflicting data in segment overlap
<code>tls.segment.toolongfragment</code>	Segment too long
<code>tls.segments</code>	Reassembled TLS segments
<code>tls.unexpected_message</code>	Unexpected message

20. RIPv2 Filters

Wireshark Filters	Description
<code>rip.auth.passwd</code>	Password
<code>rip.auth.type</code>	Authentication type
<code>rip.auth_data_len</code>	Auth Data Len
<code>rip.authentication_data</code>	Authentication Data
<code>rip.command</code>	Command

<code>rip.digest_offset</code>	Digest Offset
<code>rip.family</code>	Address Family
<code>rip.ip</code>	IP Address
<code>rip.key_id</code>	Key ID
<code>rip.metric</code>	Metric
<code>rip.netmask</code>	Netmask
<code>rip.next_hop</code>	Next Hop
<code>rip.route_tag</code>	Route Tag
<code>rip.routing_domain</code>	Routing Domain
<code>rip.seq_num</code>	Seq Number
<code>rip.unknown_address_family</code>	Unknown address family
<code>rip.zero_padding</code>	Zero adding
<code>rip.version</code>	Version

21. OSPF and OSPFv2 Filters

Wireshark Filters	Description
<code>ospf.advrouter</code>	Advertising Router
<code>ospf.area_id</code>	Area ID
<code>ospf.at</code>	OSPF Authentication Trailer
<code>ospf.at.auth_data</code>	Authentication Data
<code>ospf.at.auth_data_len</code>	Authentication Data Length
<code>ospf.at.auth_type</code>	Authentication Type

<code>ospf.at.crypto_seq_nbr</code>	Cryptographic Sequence Number
<code>ospf.at.reserved</code>	Reserved
<code>ospf.at.sa_id</code>	Security Association Identifier (SA ID)
<code>ospf.auth.crypt.data</code>	Auth Crypt Data
<code>ospf.auth.crypt.data_length</code>	Auth Crypt Data Length
<code>ospf.auth.crypt.key_id</code>	Auth Crypt Key id
<code>ospf.auth.crypt.seq_nbr</code>	Auth Crypt Sequence Number
<code>ospf.auth.none</code>	Auth Data (none)
<code>ospf.auth.simple</code>	Auth Data (Simple)
<code>ospf.auth.type</code>	Auth Type
<code>ospf.auth.unknown</code>	Auth Unknown
<code>ospf.checksum</code>	Checksum
<code>ospf.db.dd_sequence</code>	DD Sequence
<code>ospf.db.interface_mtu</code>	Interface MTU
<code>ospf.dbd</code>	DB Description
<code>ospf.dbd.i</code>	(I) Init
<code>ospf.dbd.m</code>	(M) More
<code>ospf.dbd.ms</code>	(MS) Master
<code>ospf.dbd.r</code>	(R) OOBResync
<code>ospf.dynhostname</code>	Dynamic Hostname
<code>ospf.header</code>	OSPF Header

<code>ospf.hello</code>	OSPF Hello Packet
<code>ospf.hello.active_neighbor</code>	Active Neighbor
<code>ospf.hello.backup_designated_router</code>	Backup Designated Router
<code>ospf.hello.designated_router</code>	Designated Router
<code>ospf.hello.hello_interval</code>	Hello Interval [sec]
<code>ospf.hello.interface_id</code>	Interface ID
<code>ospf.hello.network_mask</code>	Network Mask
<code>ospf.hello.router_dead_interval</code>	Router Dead Interval [sec]
<code>ospf.hello.router_priority</code>	Router Priority
<code>ospf.instance_id</code>	Instance ID
<code>ospf.link_state_id</code>	Link State ID
<code>ospf.lls.checksum</code>	Checksum
<code>ospf.lls.data_length</code>	LLS Data Length
<code>ospf.lls.ext.options</code>	Options
<code>ospf.lls.ext.options.lr</code>	(LR) LSDB Resynchronization
<code>ospf.lls.ext.options.rs</code>	(RS) Restart Signal
<code>ospf.ls.metric</code>	Metric
<code>ospf.ls.number_of_lsas</code>	Number of LSAs
<code>ospf.lsa</code>	LS Type
<code>ospf.lsa.age</code>	LS Age (seconds)
<code>ospf.mpls.routerid</code>	MPLS/TE Router ID

<code>ospf.mpls.shared_risk_link_group</code>	Shared Risk Link Group
<code>ospf.mpls.sonet.sdh</code>	SONET/SDH
<code>ospf.mpls.starting</code>	Starting n
<code>ospf.mpls.switching_type</code>	Switching Type
<code>ospf.mpls.te_metric</code>	MPLS/TE Metric
<code>ospf.mpls.type</code>	Type
<code>ospf.msg</code>	Message Type
<code>ospf.msg.dbdesc</code>	Database Description
<code>ospf.msg.hello</code>	Hello
<code>ospf.msg.lsack</code>	Link State Adv Acknowledgement
<code>ospf.msg.lsreq</code>	Link State Adv Request
<code>ospf.msg.lsupdate</code>	Link State Adv Update
<code>ospf.oif.encoding</code>	Encoding
<code>ospf.oif.local_node_id</code>	Local Node ID
<code>ospf.oif.node_id</code>	Node ID
<code>ospf.oif.remote_node_id</code>	Remote Node ID
<code>ospf.oif.signal_type</code>	Signal Type
<code>ospf.oif.switching_cap</code>	Switching Cap
<code>ospf.oif.tna_addr</code>	TNA Addr
<code>ospf.oif.tna_addr.ipv4</code>	TNA Addr
<code>ospf.oif.tna_addr.ipv6</code>	TNA Addr

<code>ospf.oif.tna_addr_length</code>	Addr Length
<code>ospf.packet_length</code>	Packet Length
<code>ospf.pad_bytes</code>	Pad Bytes
<code>ospf.prefix_length</code>	PrefixLength
<code>ospf.reserved</code>	Reserved
<code>ospf.reserved.not_zero</code>	incorrect, should be 0
<code>ospf.ri.mbz</code>	MBZ
<code>ospf.ri.options</code>	RI Options
<code>ospf.ri.options.ete</code>	(ETE) Experimental TE
<code>ospf.ri.options.grc</code>	(GRC) Graceful Restart
<code>ospf.ri.options.grh</code>	(GRH) Graceful Restart Helper
<code>ospf.ri.options.p2plan</code>	(P2PLAN) Point-to-point over LAN
<code>ospf.ri.options.srs</code>	Stub Router Support
<code>ospf.ri.options.tes</code>	(TES) Traffic Engineering
<code>ospf.tlv.adjsid.flags</code>	Flags
<code>ospf.tlv.adjsid.flags.b</code>	(B) Backup Flag
<code>ospf.tlv.adjsid.flags.g</code>	(G) Group Flag
<code>ospf.tlv.adjsid.flags.p</code>	(P) Persistent Flag
<code>ospf.tlv.adjsid.flags.unknown</code>	(*) Unknown Flag
<code>ospf.tlv.adjsid.flags.v</code>	(V) Value/Index Flag
<code>ospf.tlv.extlink.mt_id</code>	Multi-Topology ID

<code>ospf.tlv.extlink.nbr</code>	Neighbor ID
<code>ospf.tlv.extlink.subtlv_type</code>	TLV Type
<code>ospf.tlv.extlink.tlv_type</code>	TLV Type
<code>ospf.tlv.extlink.weight</code>	Weight
<code>ospf.tlv.extpfx.af</code>	Address Family
<code>ospf.tlv.extpfx.flags</code>	Flags
<code>ospf.tlv.extpfx.flags.a</code>	(A) Attach Flag
<code>ospf.tlv.extpfx.flags.n</code>	(N) Node Flag
<code>ospf.tlv.extpfx.flags.unknown</code>	(*) Unknown Flag
<code>ospf.tlv.extpfx.routetype</code>	Route Type
<code>ospf.tlv.extpfx.subtlv_type</code>	TLV Type
<code>ospf.tlv.extpfx.tlv_type</code>	TLV Type
<code>ospf.tlv.extpfx_range.flags</code>	Flags
<code>ospf.tlv.extpfx_range.flags.ia</code>	(IA) Inter-Area Flag
<code>ospf.tlv.extpfx_range.flags.unknown</code>	(*) Unknown Flag
<code>ospf.tlv.igp_msd_type</code>	MSD Type
<code>ospf.tlv.igp_msd_value</code>	MSD Value
<code>ospf.tlv.local_interface_id</code>	Local Interface ID
<code>ospf.tlv.pfxsid.flags</code>	Flags
<code>ospf.tlv.pfxsid.flags.e</code>	(E) Explicit-Null Flag
<code>ospf.tlv.pfxsid.flags.l</code>	(L) Local/Global Flag

<code>ospf.tlv.pfxsid.flags.m</code>	(M) Mapping Server Flag
<code>ospf.tlv.pfxsid.flags.np</code>	(NP) No-PHP Flag
<code>ospf.tlv.pfxsid.flags.unknown</code>	(*) Unknown Flag
<code>ospf.tlv.pfxsid.flags.v</code>	(V) Value/Index Flag
<code>ospf.tlv.preference</code>	Preference
<code>ospf.tlv.range_size</code>	Range Size
<code>ospf.tlv.remote_interface_id</code>	Remote Interface ID
<code>ospf.tlv.remote_ipv4_address</code>	Remote IPv4 Address
<code>ospf.tlv.sid_label</code>	SID/Label
<code>ospf.tlv.sidlabel_range.type</code>	TLV Type
<code>ospf.tlv.unknown</code>	Unknown TLV
<code>ospf.tlv_length</code>	TLV Length
<code>ospf.tlv_type</code>	TLV Value
<code>ospf.tlv_type.opaque</code>	TLV Value
<code>ospf.tlv_value</code>	TLV Value
<code>ospf.unknown_link_subtype</code>	Unknown Link sub-TLV
<code>ospf.unknown_text</code>	Text in the Unknown TLV
<code>ospf.srcrouter</code>	Source OSPF Router
<code>ospf.lsa.asbr</code>	Summary LSA (ASBR)
<code>ospf.lsa.asbr.netmask</code>	Netmask
<code>ospf.lsa.asext</code>	AS-External LSA (ASBR)

<code>ospf.lsa.asext.extrttag</code>	External Route Tag
<code>ospf.lsa.asext.fwdaddr</code>	Forwarding Address
<code>ospf.lsa.asext.netmask</code>	Netmask
<code>ospf.lsa.asext.type</code>	External Type
<code>ospf.lsa.attr</code>	External Attributes LSA
<code>ospf.lsa.bc_error</code>	BC error
<code>ospf.lsa.chksum</code>	Checksum
<code>ospf.lsa.donotage</code>	Do Not Age Flag
<code>ospf.lsa.id</code>	Link State ID
<code>ospf.lsa.invalid_length</code>	Invalid length
<code>ospf.lsa.length</code>	Length
<code>ospf.lsa.member</code>	Group Membership LSA
<code>ospf.lsa.mpls</code>	MPLS Traffic Engineering LSA
<code>ospf.lsa.network</code>	Network LSA
<code>ospf.lsa.network.attchrtr</code>	Attached Router
<code>ospf.lsa.network.netmask</code>	Netmask
<code>ospf.lsa.nssa</code>	NSSA AS-External LSA
<code>ospf.lsa.number_of_links</code>	Number of Links
<code>ospf.lsa.opaque</code>	Opaque LSA
<code>ospf.lsa.router</code>	Router LSA
<code>ospf.lsa.router.linkdata</code>	Link Data

<code>ospf.lsa.router.linkid</code>	Link ID
<code>ospf.lsa.router.linktype</code>	Link Type
<code>ospf.lsa.router.metric0</code>	0 Metric
<code>ospf.lsa.router.nummetrics</code>	Number of Metrics
<code>ospf.lsa.seqnum</code>	Sequence Number
<code>ospf.lsa.summary</code>	Summary LSA (IP Network)
<code>ospf.lsa.tos</code>	TOS
<code>ospf.lsa.tos_missing</code>	Blocks missing
<code>ospf.lsa.unknown_type</code>	Unknown LSA Type
<code>ospf.lsa_sa</code>	SR-Algorithm
<code>ospf.lsid.opaque_id</code>	Link State ID Opaque ID
<code>ospf.lsid_opaque_type</code>	Link State ID Opaque Type
<code>ospf.lsid_te_lsa.instance</code>	Link State ID TE-LSA Instance
<code>ospf.lsid_te_lsa.reserved</code>	Link State ID TE-LSA Reserved
<code>ospf.metric</code>	Metric
<code>ospf.mpls.action</code>	Action
<code>ospf.mpls.bandwidth.type</code>	Bandwidth Type
<code>ospf.mpls.bc</code>	BC
<code>ospf.mpls.bc.model_id</code>	MPLS/DSTE Bandwidth Constraints Model Id
<code>ospf.mpls.bitmap</code>	Bitmap
<code>ospf.mpls.cs</code>	Channel Spacing

<code>ospf.mpls.effective</code>	No. of effective. Bits
<code>ospf.mpls.encoding</code>	Encoding
<code>ospf.mpls.grid</code>	Grid
<code>ospf.mpls.group</code>	MPLS/TE Group
<code>ospf.mpls.interface_mtu</code>	Interface MTU
<code>ospf.mpls.length</code>	Length
<code>ospf.mpls.link_max_bw</code>	Link Max BW
<code>ospf.mpls.linkcolor</code>	MPLS/TE Link Resource Class/Color
<code>ospf.mpls.linkid</code>	MPLS/TE Link ID
<code>ospf.mpls.linktype</code>	MPLS/TE Link Type
<code>ospf.mpls.local_addr</code>	MPLS/TE Local Interface Address
<code>ospf.mpls.local_id</code>	MPLS/TE Local Interface Index
<code>ospf.mpls.minimum_lsp_bandwidth</code>	Minimum LSP bandwidth
<code>ospf.mpls.n</code>	Starting n
<code>ospf.mpls.num.labels</code>	Num Labels
<code>ospf.mpls.pri</code>	Pri (or TE-Class)
<code>ospf.mpls.priority</code>	Priority
<code>ospf.mpls.protection_capability</code>	Protection Capability
<code>ospf.mpls.remote_addr</code>	MPLS/TE Remote Interface Address
<code>ospf.mpls.remote_id</code>	MPLS/TE Remote Interface Index
<code>ospf.v2.grace</code>	Grace TLV

<code>ospf.v2.grace.ip</code>	Restart IP
<code>ospf.v2.grace.period</code>	Grace Period
<code>ospf.v2.grace.reason</code>	Restart Reason
<code>ospf.v2.lls.auth_data</code>	Auth Data
<code>ospf.v2.lls.sequence_number</code>	Sequence number
<code>ospf.v2.options</code>	Options
<code>ospf.v2.options.dc</code>	(DC) Demand Circuits
<code>ospf.v2.options.dn</code>	(E) External Routing
<code>ospf.v2.options.l</code>	(L) LLS Data block
<code>ospf.v2.options.mc</code>	(MC) Multicast
<code>ospf.v2.options.mt</code>	(MT) Multi-Topology Routing
<code>ospf.v2.options.n</code>	(N) NSSA
<code>ospf.v2.options.p</code>	(P) Propagate
<code>ospf.v2.router.lsa.flags</code>	Flags
<code>ospf.v2.router.lsa.flags.b</code>	(B) Area border router
<code>ospf.v2.router.lsa.flags.e</code>	(E) AS boundary router
<code>ospf.v2.router.lsa.flags.n</code>	(N) flag
<code>ospf.v2.router.lsa.flags.v</code>	(V) Virtual link endpoint
<code>ospf.v2.router.lsa.flags.w</code>	(W) Wild-card multicast receiver

22. OSPFv3/IP v6 Filters

Wireshark Filters	Description
<code>ospf.v3.as.external.flags</code>	Flags
<code>ospf.v3.as.external.flags.e</code>	(E) External Metric
<code>ospf.v3.as.external.flags.f</code>	(F) Forwarding Address
<code>ospf.v3.as.external.flags.t</code>	(T) External Route Tag
<code>ospf.v3.address_prefix.ipv4</code>	Address Prefix
<code>ospf.v3.address_prefix.ipv6</code>	Address Prefix
<code>ospf.v3.lls.dropped_neighbor</code>	Dropped Neighbor
<code>ospf.v3.lls.drop.tlv</code>	Neighbor Drop TLV
<code>ospf.v3.lls.ext.options</code>	Options
<code>ospf.v3.lls.ext.options.lr</code>	(LR) LSDB Resynchronization
<code>ospf.v3.lls.ext.options.rs</code>	(RS) Restart Signal
<code>ospf.v3.lls.willingness.tlv</code>	Extended Options TLV, Willingness TLV
<code>ospf.v3.lls.extended_options</code>	Extended Options
<code>ospf.v3.options</code>	Options
<code>ospf.v3.lls.ext.options.tlv</code>	Extended Options TLV
<code>ospf.v3.lls.fsf.tlv</code>	Full State For TLV
<code>ospf.v3.lls.full_state_for</code>	Full State For
<code>ospf.v3.lls.ll_id</code>	Local Interface ID
<code>ospf.v3.lls.neighbor</code>	Neighbor

<code>ospf.v3.lls.relay.added</code>	Relays Added
<code>ospf.v3.lls.relay.options</code>	Options
<code>ospf.v3.lls.relay.options.a</code>	(A) Always
<code>ospf.v3.lls.relay.options.n</code>	(N) Never
<code>ospf.v3.lls.relay.tlv</code>	Active Overlapping Relays TLV
<code>ospf.v3.lls.request_from</code>	Request From
<code>ospf.v3.lls.rf.tlv</code>	Request From TLV
<code>ospf.v3.lls.state.options</code>	Options
<code>ospf.v3.lls.state.options.a</code>	(A) Answer
<code>ospf.v3.lls.state.options.n</code>	(N) Incomplete
<code>ospf.v3.lls.state.options.r</code>	(R) Resuest
<code>ospf.v3.lls.state.scs</code>	SCS Number
<code>ospf.v3.lls.state.tlv</code>	State Check Sequence TLV
<code>ospf.v3.lls.willingness</code>	Willingness
<code>ospf.v3.lsa</code>	LS Type
<code>ospf.v3.lsa.asext</code>	AS-External-LSA
<code>ospf.v3.lsa.attached_router</code>	Attached Router
<code>ospf.v3.lsa.destination_router_id</code>	Destination Router ID
<code>ospf.v3.lsa.do_not_age</code>	Do Not Age
<code>ospf.v3.lsa.external_route_tag</code>	External Route Tag
<code>ospf.v3.lsa.fc</code>	Function Code

<code>ospf.v3.lsa.forwarding_address.ipv4</code>	Forwarding Address
<code>ospf.v3.lsa.forwarding_address.ipv6</code>	Forwarding Address
<code>ospf.v3.lsa.interface_id</code>	Interface ID
<code>ospf.v3.lsa.interprefix</code>	Inter-Area-Prefix-LSA
<code>ospf.v3.lsa.interrouter</code>	Inter-Area-Router-LSA
<code>ospf.v3.lsa.intraprefix</code>	Intra-Area-Prefix-LSA
<code>ospf.v3.lsa.link</code>	Link-LSA
<code>ospf.v3.lsa.link_local_interface_address.ipv4</code>	Link-local Interface Address
<code>ospf.v3.lsa.link_local_interface_address.ipv6</code>	Link-local Interface Address
<code>ospf.v3.lsa.member</code>	Group-Membership-LSA
<code>ospf.v3.lsa.neighbor_interface_id</code>	Neighbor Interface ID
<code>ospf.v3.lsa.neighbor_router_id</code>	Neighbor Router ID
<code>ospf.v3.lsa.network</code>	Network-LSA
<code>ospf.v3.lsa.nssa</code>	NSSA-LSA
<code>ospf.v3.lsa.num_prefixes</code>	# prefixes
<code>ospf.v3.lsa.opaque</code>	Router Information Opaque-LSA
<code>ospf.v3.lsa.referenced_advertising_router</code>	Referenced Advertising Router
<code>ospf.v3.lsa.referenced_link_state_id</code>	Referenced Link State ID
<code>ospf.v3.lsa.referenced_ls_type</code>	Referenced LS type
<code>ospf.v3.lsa.router</code>	Router-LSA
<code>ospf.v3.lsa.router_priority</code>	Router Priority

<code>ospf.v3.lsa.s12</code>	Flooding Scope
<code>ospf.v3.lsa.type</code>	Type
<code>ospf.v3.lsa.u</code>	LSA Handling
<code>ospf.v3.prefix.options</code>	PrefixOptions
<code>ospf.v3.prefix.options.la</code>	(LA) Local Address
<code>ospf.v3.prefix.options.mc</code>	(MC) Multicast
<code>ospf.v3.prefix.options.nu</code>	(NU) NoUnicast
<code>ospf.v3.prefix.options.p</code>	(P) Propagate
<code>ospf.v3.router.lsa.flags</code>	Flags
<code>ospf.v3.router.lsa.flags.b</code>	(B) Area border router
<code>ospf.v3.router.lsa.flags.e</code>	(E) AS boundary router
<code>ospf.v3.router.lsa.flags.v</code>	(V) Virtual link endpoint
<code>ospf.v3.router.lsa.flags.w</code>	(W) Wild-card multicast receiver
<code>ospf.version</code>	Version

23. Common Filtering Commands

Wireshark Filters	Description
<code>ip.addr == 10.10.50.1/24</code>	Filter subnet: display traffic from the subnet, be its source or destination
<code>tcp.port == 25</code>	Filter by TCP port
<code>tcp.dstport == 23</code>	Filter by destination TCP port

<code>ip.addr == 10.10.50.1 and Tcp.port == 25</code>	Filter by IP address and port
<code>http.host == "host name"</code>	Filter by URL
<code>frame.time >= "June 02, 2019 18:04:00"</code>	Filter by timestamp
<code>tcp.flags.syn == 1 or (tcp.seq==1 and tcp.ack==1 and tcp.len==0 and tcp.analysis.initial_rtt)</code>	Filter SYN flag; Filter three-way handshake
<code>tcp.flags.syn == 1 and tcp.flags.ack == 0</code>	Filter SYN flag; Detecting SYN Floods
<code>wlan.fc.type_subtype = 0x08</code>	Wireshark Beacon Filter
<code>eth.dst == ff:ff:ff:ff:ff:ff</code>	Wireshark broadcast filter
<code>(eth.dst[0] & 1)</code>	Wireshark Multicast filter
<code>ip.host == (hostname)</code>	Host name filter
<code>eth.addr == 00:70:f4:23:18:c4</code>	MAC address filter
<code>tcp.flags.reset == 1</code>	RST flag filter
<code>ip.addr != 192.168.0.1</code>	Exclude IP address: remove traffic from and to IP address
<code>frame contains "(attachment tar exe zip pdf)"</code>	Find files by type
<code>tcp contains <Keyword> frame contains <Keyword></code>	Find traffic based on keyword

24. Filtering Packets

Operator	Description
<code>eq or ==</code>	Equal <code>ip.dest == 192.168.1.1</code>
<code>ne or !=</code>	Not Equal <code>ip.dest != 192.168.1.1</code>

<code>gt or ></code>	Greater than frame.len > 10
<code>lt or <</code>	Less than frame.len <10
<code>ge or >=</code>	Greater than or equal frame.len >= 10
<code>le or <=</code>	Less than or equal frame.len<=10

25. Logical Operators

Operators	Description
<code>and or &&</code>	Logical AND All the conditions should match
<code>or or </code>	Logical OR Either all or one of the conditions should match
<code>xor or ^^</code>	Logical XOR exclusive alternation – Only one of the two conditions should match not both
<code>not or !</code>	Logical NOT Not equal to
<code>[n] [...]</code>	Substring operator Filter a specific word or text

26. Default Columns in Packet Capture Output

Columns	Description
<code>No.</code>	Frame number from the beginning of the packet capture
<code>Time</code>	Seconds from the first frame
<code>Source (src)</code>	Source address, commonly an IPv4, IPv6 or Ethernet address
<code>Destination (dst)</code>	Destination address

Protocol	The protocol used in the Ethernet frame, IP packet, or TCP segment
Length	Length of the frame in bytes

27. Display Window Keyboard Shortcuts

Accelerator	Description
Tab or Shift+Tab	Move between screen elements, e.g. from the toolbars to the packet list to the packet detail
Alt+→ or Option+→	Move to the next packet in the selection history
↓	Move to the next packet or detail item
↑	Move to the previous packet or detail item
→	In the packet detail, opens the selected tree item
Shift+→	In the packet detail, opens the selected tree item and all its subtrees
Ctrl+ ↓ or F8	Move to the next packet, even if the packet list isn't focused
Ctrl+ ↑ or F7	Move to the previous packet, even if the packet list isn't focused
Ctrl+←	In the packet detail, closes all tree items
Ctrl+→	In the packet detail, opens all tree items
Ctrl+.	Move to the next packet of the conversation (TCP, UDP or IP)
Ctrl+,	Move to the previous packet of the conversation (TCP, UDP or IP)
Backspace	In the packet detail, jumps to the parent node
Return or Enter	In the packet detail, toggles the selected tree item

28. Other Commands

Commands	Description
<code>tftp irc bittorrent</code>	Displays packets that match any of these three protocols.
<code>frame[100-199] contains "nessus"</code>	Frame offset 100-199 contains

	"nessus" in uc/lc
<code>http.user_agent contains "nmap"</code>	Suspected nmap traffic (case sensitive)
<code>frame matches "join #"</code>	IRC Joins
<code>ftp.request.command=="USER" && tcp.len>50</code>	Long FTP Username